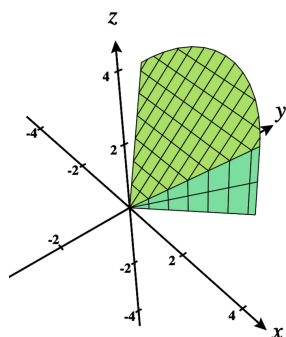


In Class Assignment Part I - Setting Up Integrals

Group Member Names :

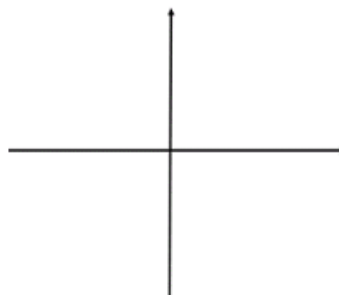
A. Yellow

For the yellow solid, in the orientation pictured below, write the integral that finds the mass of the solid, given that the mass density function of the solid is $\rho(x, y) = x^2 + y^2$. Fully set up the integral both in rectangular coordinates and in cylindrical coordinates. Draw the shadow of the solid on the axes provided. The surfaces bounding the solid are $y = x$, $y = -x$, $z = y$, and $x^2 + y^2 = 9$.



Rectangular Coordinates

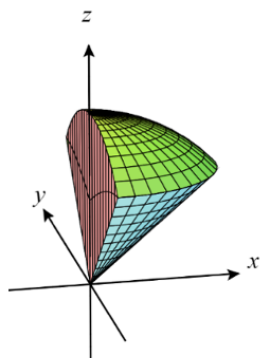
Shadow Region



Cylindrical Coordinates

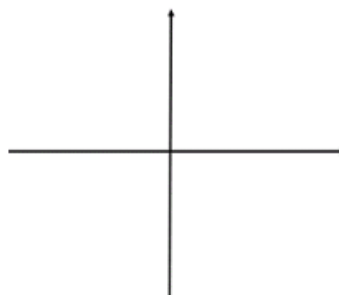
B. Purple

For the purple solid, in the orientation pictured below, fully set up the triple integral that finds the volume of the solid in cylindrical coordinates and in spherical coordinates. Draw the shadow of the solid on the axes provided. Note that the shadow region only applies to the cylindrical coordinates. Assume that the sphere surface has radius $\sqrt{2}$.



Cylindrical Coordinates

Shadow Region



Spherical Coordinates

In Class Assignment Part II - Matching Solids to Their Integrals

1. Identify the coordinate system used for each integral.
2. If the integral is cartesian or cylindrical, draw the shadow region.
3. Match the 3D solid with the integral that computes it's volume. Note that radius values are not to scale.

Integral	Coordinate System	Shadow Region	Color of the Solid
1. $\int_0^{2\pi} \int_3^5 \int_0^{\sqrt{25-r^2}} r \, dz \, dr \, d\theta$			
2. $\int_{-4}^4 \int_0^{\sqrt{16-x^2}} \int_0^y 1 \, dz \, dy \, dx$			
3. $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \int_0^5 \int_{-5}^{-r} r \, dz \, dr \, d\theta$			
4. $\int_0^4 \int_{-\sqrt{16-y^2}}^{\sqrt{16-y^2}} y \, dx \, dy$			
5. $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \int_0^{2\pi} \int_0^4 \rho^2 \sin \phi \, d\rho \, d\theta \, d\phi$			
6. $\int_0^{\frac{\pi}{4}} \int_{\pi}^{2\pi} \int_0^{\sqrt{2}} \rho^2 \sin \phi \, d\rho \, d\theta \, d\phi$			
7. $\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \int_0^{\pi} \int_3^5 \rho^2 \sin \phi \, d\rho \, d\theta \, d\phi$			